

Linear differential equations.

Let $m \in \mathbb{N}^*$. $I \subset \mathbb{R}$ open interval.

$a_1, \dots, a_m \in C(I)$.

↳ space of all continuous functions.

$C(I), C'(I), \dots, C^m(I)$.

$\mathcal{L}: C^m(I) \rightarrow C(I)$.

$x \in C^m(I) \mapsto \mathcal{L}(x) \in C(I)$.

$$\mathcal{L}(x)(t) = x^{(m)}(t) + a_1(t) \cdot x^{(m-1)}(t) + \dots + a_{m-1}(t) \cdot x'(t) + a_m(t) \cdot x(t).$$

We consider the m^{th} order LHDE: $\mathcal{L}(x) = 0$.

Lemma: The map $\mathcal{L}$ is linear, $\mathcal{L}(\alpha x + \beta y) = \alpha \cdot \mathcal{L}(x) + \beta \cdot \mathcal{L}(y)$,
 $(\forall) x, y \in C^m(I)$, $(\forall) \alpha, \beta \in \mathbb{R}$.
 (functions).

→ Consequently, the set of all solutions of the LHDE (1) is denoted by
Ker $\mathcal{L}$ (the linear space).

Proof.

$$\mathcal{L}(\alpha x + \beta y) = (\alpha x + \beta y)^{(m)} + a_1(\alpha x + \beta y)^{(m-1)} + \dots + a_{m-1}(\alpha x + \beta y)' + a_m(\alpha x + \beta y).$$

$$(\alpha x + \beta y)' = \alpha \cdot x' + \beta \cdot y' \text{ (linearity of the first order derivative).}$$

$$\begin{aligned} \mathcal{L}(\alpha x + \beta y) &= \alpha(x^{(m)} + \dots + a_m \cdot x) + \beta(y^{(m)} + \dots + a_m \cdot y) \\ &= \alpha \cdot \mathcal{L}(x) + \beta \cdot \mathcal{L}(y). \end{aligned}$$

! Kernel of a linear map is a linear space.

Theorem: (Fundamental theorem for LHDE).

The dimension of the $\text{Ker } \mathcal{L}$ is m .
 (because it has a base with m elements).

↳ set of lin. ind. elems.

• More precisely, $\exists x_1, \dots, x_m$ lin. ind. elements in the $\text{Ker } \mathcal{L}$ s.t.

$$\text{Ker } \mathcal{L} = \langle x_1, \dots, x_m \rangle = \{ c_1 \cdot x_1 + c_2 \cdot x_2 + \dots + c_m \cdot x_m \mid c_1, \dots, c_m \in \mathbb{R} \}.$$

(span)

• In other words, the DE has m lin. ind. solutions $x_1, \dots, x_m$ and the general solution of (1) is $x = c_1 \cdot x_1 + c_2 \cdot x_2 + \dots + c_m \cdot x_m$; $c_1, \dots, c_m \in \mathbb{R}$.
 ↳ "fundamental set of solutions".

In other terms, cht and sht are lin. ind. sol.

$$x = c_1 \cdot cht + c_2 \cdot sht, \quad c_1, c_2 \in \mathbb{R}.$$

EX: Look for $\lambda \in \mathbb{R}$ s.t. $x = e^{\lambda t}$ is a sol. of (2).

$$x = e^{\lambda t}$$

$$x' = \lambda \cdot e^{\lambda t}$$

$\vdots$

$$x^{(k)} = \lambda^k \cdot e^{\lambda t}$$

$$\mathcal{L}(e^{\lambda t}) = e^{\lambda t} (l(\lambda)), \quad (\lambda) t \in \mathbb{R}.$$

common factor.

$$l(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n \quad (3).$$

make notation $\rightarrow$ polynomial of degree n with those coeff.

$e^{\lambda t}$ is a solution of $\mathcal{L}(x) = 0$

$$\Leftrightarrow \mathcal{L}(e^{\lambda t}) = 0, \quad (\lambda) t \in \mathbb{R}.$$

$$\Leftrightarrow e^{\lambda t} \cdot l(\lambda) = 0$$

$$\Leftrightarrow l(\lambda) = 0$$

$\Rightarrow \lambda$ is a root of the polynomial l .

Def: The polynomial (3) is called the characteristic polynomial of LHE with $cc(2)$; while $l(\lambda) = 0$ is the characteristic equation.

Exercise: Find characteristic polynomial of $x' + ax = 0$

$$x' + ax = 0 \Rightarrow \lambda + a = 0 \Rightarrow \lambda = -a \quad \text{one root only!}$$

$$x'' + a_1 x' + a_2 x = 0 \Rightarrow \lambda^2 + a_1 \lambda + a_2 = 0.$$

$$\lambda = -a \rightarrow e^{-at} \text{ a solution.}$$

(associate a function $\rightarrow$ that function is a solution).

$$x = c \cdot e^{-at}, \quad c \in \mathbb{R}.$$

$$x'' - 5x' + 6x = 0 \Rightarrow \lambda^2 - 5\lambda + 6 = 0$$

($m=2$ order)

$$\lambda_1 = 2 \rightarrow e^{2t}$$

$$\lambda_2 = 3 \rightarrow e^{3t}$$

$$x = c_1 \cdot e^{2t} + c_2 \cdot e^{3t}$$

$$c_1, c_2 \in \mathbb{R}$$

Proof: $\dim \ker \mathcal{L} = m$ (ex: $\mathbb{R}^m$ lin. space of dim m). 2nd step

- We achieve this by proving that $\ker \mathcal{L}$ is isomorphic to $\mathbb{R}^m$.
- Recall the existence and uniqueness theorem:

for a fixed $t_0 \in I$ and $(*) \eta \in \mathbb{R}^m$, the IVP
$$\begin{cases} \mathcal{L}(x) = 0 \\ x(t_0) = \eta_1 \\ x'(t_0) = \eta_2 \\ \vdots \\ x^{(m-1)}(t_0) = \eta_m \end{cases}$$
 has a unique sol. $p \in C^m(I)$.

$\Rightarrow$ this is the bijective correspondence that we need.

- This statement assures that the map: ...

$$T: \ker \mathcal{L} \rightarrow \mathbb{R}^m, T(p) = (p(t_0), p'(t_0), \dots, p^{(m-1)}(t_0)), (*) p \in \ker \mathcal{L}.$$

... can be rewritten like this: $\uparrow$

$$(*) \eta \in \mathbb{R}^m, \exists! p \in C^m(I) \text{ s.t. } \begin{cases} \mathcal{L}(p) = 0 \\ T(p) = \eta \end{cases}$$

(there is a unique function p that verifies these conditions).

$$\Leftrightarrow (*) \eta \in \mathbb{R}^m, \exists! p \in \ker \mathcal{L} \text{ s.t. } T(p) = \eta \Leftrightarrow T \text{ bijective (1).}$$

$$T \text{ is linear } (\Rightarrow) T(\alpha p + \beta \psi) = \alpha \cdot T(p) + \beta \cdot T(\psi), (*) p, \psi \in \ker \mathcal{L}, \alpha, \beta \in \mathbb{R}.$$

$$\begin{aligned} \Leftrightarrow T(\alpha p + \beta \psi) &= \begin{pmatrix} \alpha \cdot p(t_0) + \beta \cdot \psi(t_0) \\ \alpha \cdot p'(t_0) + \beta \cdot \psi'(t_0) \\ \vdots \\ \alpha \cdot p^{(m-1)}(t_0) + \beta \cdot \psi^{(m-1)}(t_0) \end{pmatrix} \\ &= \alpha \cdot \begin{pmatrix} p(t_0) \\ p'(t_0) \\ \vdots \\ p^{(m-1)}(t_0) \end{pmatrix} + \beta \cdot \begin{pmatrix} \psi(t_0) \\ \psi'(t_0) \\ \vdots \\ \psi^{(m-1)}(t_0) \end{pmatrix} = \alpha \cdot T(p) + \beta \cdot T(\psi) \quad (2). \end{aligned}$$

(1), (2) $\Rightarrow$ T linear and bijective $\Rightarrow T$ is an isomorphism between $\ker \mathcal{L}$ and $\mathbb{R}^m$.

Questions: (m) with 20 marks

m = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

Proof

- Are there methods to find solutions?
- And afterwards to prove they are lin. indep.?

LHDE with CC

↳ y constant coeff.

Let $m \in \mathbb{N}^*$, $a_1, \dots, a_m \in \mathbb{R}$.

$$L: C^m(\mathbb{R}) \rightarrow C(\mathbb{R})$$

$$x \in C^m(\mathbb{R}) \mapsto L(x) \in C(\mathbb{R})$$

$$L(x) = x^{(m)} + a_1 x^{(m-1)} + \dots + a_{m-1} x' + a_m x$$

$$L(x) = 0 \quad (2)$$

- Aim: to present an algorithm that finally gives m lin. indep. solutions of (2).

Ex: 1) $x' - x = 0$.

We need 1 non-null solution.

$$x_1 = e^t$$

The general solution is:

$$x = c \cdot e^t, c \in \mathbb{R} \quad (\text{using theorem 1})$$

2) $x' + ax = 0$, $a \in \mathbb{R}$ fixed.

(2) for $m=1$.

$$x = e^{-at} \Rightarrow x' = -a \cdot e^{-at}$$

$$\Rightarrow -a \cdot e^{-at} + a \cdot e^{-at} = 0, (\forall) t \in \mathbb{R}.$$

The general solution is:

$$x = c \cdot e^{-at}, c \in \mathbb{R}.$$

3) $m=2$: $x'' + a_1 x' + a_2 x = 0$ (order)

$$x'' - x = 0$$

$$x_1 = e^t, x_2 = e^{-t}.$$

$\{x_1, x_2\}$ are lin. indep.

The general solution:

$$x = c_1 \cdot e^t + c_2 \cdot e^{-t}, c_1, c_2 \in \mathbb{R}$$

degree n ?

The roots of the polynomial of degree n . (3)

TFA. ℓ has n roots in $\mathbb{C}$, counted with their corresponding multiplicities.

Let $x_1, \dots, x_m \in \mathbb{C}$ be those roots (not necessarily distinct).

$$\Rightarrow \ell(x) = (x - x_1)(x - x_2) \cdots (x - x_m).$$

$$\bullet x^2 - 2x + 1 = 0$$

$$\Rightarrow (x-1)^2 = 0 \Rightarrow 1 \text{ is a double root.}$$

$$\bullet x^2 + 9 = 0$$

$$x_1 = -3i \in \mathbb{C}$$

$$x_2 = 3i \in \mathbb{C}.$$

Complex solutions of the LHDE (2).

Let $u, v \in C^{(n)}(\mathbb{R})$ and $\ell: \mathbb{R} \rightarrow \mathbb{C}$ be $\ell(t) = u(t) + i \cdot v(t)$, $t \in \mathbb{R}$.

$$\ell' = u' + i \cdot v'$$

$$\ell'' = u'' + i \cdot v''$$

If $\mathcal{L}(\ell) = 0$ then we say that ℓ is a complex solution of $\mathcal{L}(x) = 0$.

Prop: If $\ell: \mathbb{R} \rightarrow \mathbb{C}$, $\ell = u + i \cdot v$ (where $u, v: \mathbb{R} \rightarrow \mathbb{R}$) is a complex solution of $\mathcal{L}(x) = 0$, then both u and v are (real) solutions of the same diff equation.

Proof:

$$\mathcal{L}(\ell) = \mathcal{L}(u + i \cdot v) = \underbrace{\mathcal{L}(u)}_{\text{real}} + i \cdot \underbrace{\mathcal{L}(v)}_{\text{real}}.$$

$$\text{hyp } (\Rightarrow) \mathcal{L}(\ell) = 0$$

$$\Rightarrow \mathcal{L}(u) + i \cdot \mathcal{L}(v) = 0$$

$$\Rightarrow \mathcal{L}(u) = \mathcal{L}(v) = 0.$$

Remark: $\lambda \in \mathbb{C}$ is a root of $\ell \Leftrightarrow e^{\lambda t}$ is a (complete) solution of $\mathcal{L}(x) = 0$

$e^{it} \rightarrow$ complex function

$$e^{(1+i) \cdot t}.$$

$$h = \alpha + i \cdot \beta, \quad \alpha, \beta \in \mathbb{R}.$$

$\Rightarrow$ we need the real / imaginary part of $e^{(\alpha + i \cdot \beta)t} = u + i \cdot v$
 $= u(t) + i \cdot v(t)$. ATT

The Euler's formula:

$$e^{\alpha + i \cdot \beta} = e^{\alpha} \cdot e^{i \cdot \beta} = e^{\alpha} \cdot (\cos \beta + i \cdot \sin \beta).$$

$$e^{(\alpha + i \cdot \beta) \cdot t} = e^{\alpha t} \cdot (\cos \beta t + i \cdot \sin \beta t), \quad t \in \mathbb{R}.$$

$$\alpha = 0, \quad \beta = \pi:$$

$$e^{i\pi} = \cos \pi + i \cdot \sin \pi = -1.$$

$$\Rightarrow \boxed{e^{i\pi} + 1 = 0} \quad \text{YT "the most beautiful formula of maths".}$$

Proof: If $\lambda \in \mathbb{C}$ is a complex solution of $\chi(t) = 0$ then we say that λ is a complex solution of $\chi(x) = 0$.

Prop: If $\lambda \in \mathbb{C}$ is a complex solution of $\chi(x) = 0$ then $\bar{\lambda}$ is also a solution of $\chi(x) = 0$.
 Proof: $\chi(\lambda) = 0 \Rightarrow \chi(\bar{\lambda}) = 0$.

$$\chi(\lambda) = 0 \Rightarrow \chi(\bar{\lambda}) = 0$$

$$\begin{aligned} 0 &= \chi(\lambda) = 0 \\ 0 &= \chi(\bar{\lambda}) = 0 \end{aligned}$$

Remark: $\lambda \in \mathbb{C}$ is a root of χ if and only if $\bar{\lambda}$ is a root of χ .

$$e^{i\pi} = -1$$